

Algebraic structures and group preliminaries

Rings, fields, vector spaces, subgroups, and cyclic groups

Applied Algebra
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Where we are, and where we are going

Last lecture focused on groups:

$$(\mathbb{Z}, +), \quad (\mathbb{Z}_n, +), \quad G_n, \quad S_n, \quad A_n, \quad D_n.$$

Today we do two things.

Block	Topic
1	Rings and fields: two operations, zero-divisors, $\mathbb{Z}_p$.
2	Vector spaces: fields acting on Abelian groups.
3	Group preliminaries: cancellation, powers, subgroups, cyclic groups.

Rings and fields

From groups to rings

A group has **one operation**. A ring has **two operations**, usually

$$+ \quad \text{and} \quad \cdot.$$

The motivating example is

$$(\mathbb{Z}, +, \cdot).$$

Addition behaves like an Abelian group:

$$0, \quad -a, \quad a + b = b + a.$$

Multiplication is associative and distributes over addition:

$$a(b + c) = ab + ac, \quad (a + b)c = ac + bc.$$

Definition of a ring

A **ring** is a set R with two operations $+$ and $\cdot$ such that:

- 1 $(R, +)$ is an Abelian group; in particular, it has an additive identity 0 ;
- 2 multiplication is closed and associative;
- 3 multiplication distributes over addition:

$$x(y + z) = xy + xz, \quad (x + y)z = xz + yz.$$

Useful vocabulary:

- an **identity element for multiplication** is an element $1 \in R$ such that

$$1a = a1 = a \quad \text{for all } a \in R;$$

a ring may or may not have such an element;

- a ring is **commutative** if $ab = ba$ for all $a, b \in R$.

Example: $\mathbb{Z}$ and $2\mathbb{Z}$

Example 1.

$$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$$

is a commutative ring with identity 1.

Example 2.

$$2\mathbb{Z} = \{2k : k \in \mathbb{Z}\}$$

is also a ring under the usual operations.

But $2\mathbb{Z}$ has **no multiplicative identity**.

So a ring need not have all the familiar properties of $\mathbb{Z}$.

Example: matrix rings

Let

$$M_2(\mathbb{R}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a, b, c, d \in \mathbb{R} \right\}.$$

Under matrix addition and multiplication, $M_2(\mathbb{R})$ is a ring with identity

$$I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

But it is not commutative in general, and it has zero-divisors:

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

$\mathbb{Z}_n$ as a ring

For $n \geq 2$, the congruence classes modulo n form a ring:

$$\mathbb{Z}_n = \{[0]_n, [1]_n, \dots, [n-1]_n\}.$$

Operations are defined modulo n :

$$[a]_n + [b]_n = [a + b]_n, \quad [a]_n [b]_n = [ab]_n.$$

A useful ring consequence

Theorem. Let R be any ring and let $x \in R$. Then

$$x0 = 0 = 0x.$$

Proof idea. Let $y = x0$. Then

$$y + y = x0 + x0 = x(0 + 0) = x0 = y.$$

Add $-y$ to both sides:

$$y = 0.$$

Thus $x0 = 0$. The proof of $0x = 0$ is similar.

Abstract algebra often works like this: axioms force useful consequences.

Definition of a field

A **field** is a commutative ring F with identity $1 \neq 0$ such that every non-zero element has a multiplicative inverse.

Equivalently, in a field we can add, subtract, multiply, and divide by non-zero elements.

Examples.

$\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, $\mathbb{Z}_p$ when p is prime.

Non-examples.

$\mathbb{Z}$, $\mathbb{Z}_6$, $M_2(\mathbb{R})$, $\mathbb{Z}_n$ when n is composite.

Example: inverses in $\mathbb{Z}_5$

In $\mathbb{Z}_5$, every non-zero element has an inverse:

a	1	2	3	4
a^{-1}	1	3	2	4

Check:

$$2 \cdot 3 = 6 \equiv 1 \pmod{5}, \quad 4 \cdot 4 = 16 \equiv 1 \pmod{5}.$$

In contrast, $\mathbb{Z}_6$ is not a field because $[2]_6$ has no multiplicative inverse.

A slightly larger field: $\mathbb{Q}[\sqrt{2}]$

Define

$$\mathbb{Q}[\sqrt{2}] = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\}.$$

This is a field. The main point is inverses. If $a + b\sqrt{2} \neq 0$, then

$$(a + b\sqrt{2})^{-1} = \frac{a - b\sqrt{2}}{a^2 - 2b^2}.$$

Indeed,

$$(a + b\sqrt{2})(a - b\sqrt{2}) = a^2 - 2b^2.$$

This illustrates enlarging a field by adjoining a new number.

Quick check: rings and fields

Decide whether each object is a ring, a field, both, or neither.

- 1 $\mathbb{Z}$ under usual $+$ and $\cdot$.
- 2 $2\mathbb{Z}$ under usual $+$ and $\cdot$.
- 3 $\mathbb{Z}_5$ under modular $+$ and $\cdot$.
- 4 $\mathbb{Z}_6$ under modular $+$ and $\cdot$.
- 5 $M_2(\mathbb{R})$ under matrix $+$ and $\cdot$.
- 6 $\mathbb{Q}[\sqrt{2}]$ under usual $+$ and $\cdot$.

For each non-field, identify what fails.

Vector spaces

A first example: $\mathbb{R}^2$

Let

$$\mathbb{R}^2 = \{(x, y) : x, y \in \mathbb{R}\}.$$

We can add elements coordinatewise:

$$(x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2).$$

We can also multiply an element by a real number:

$$\lambda(x, y) = (\lambda x, \lambda y).$$

For example,

$$2(3, -1) = (6, -2).$$

Elements of $\mathbb{R}^2$ are called **vectors**; elements of $\mathbb{R}$ are called **scalars**.

What structure does $\mathbb{R}^2$ have?

The set $\mathbb{R}^2$ has two kinds of operations.

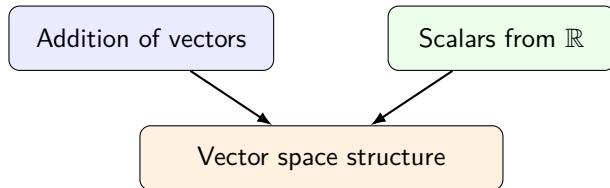
- We can **add vectors**:

$$u + v \in \mathbb{R}^2.$$

- We can **scale vectors** by real numbers:

$$\lambda v \in \mathbb{R}^2.$$

So $\mathbb{R}^2$ combines:



Definition of a vector space

Let F be a field.

A **vector space** over F is an additive Abelian group $(V, +)$ together with scalar multiplication

$$F \times V \rightarrow V, \quad (\lambda, v) \mapsto \lambda v,$$

satisfying:

$$\begin{aligned} (\lambda\mu)v &= \lambda(\mu v), & 1v &= v, \\ (\lambda + \mu)v &= \lambda v + \mu v, & \lambda(u + v) &= \lambda u + \lambda v. \end{aligned}$$

Elements of V are **vectors**; elements of F are **scalars**.

Examples of vector spaces

The example $\mathbb{R}^2$ is a vector space over $\mathbb{R}$.

More generally,

$$\mathbb{R}^n = \{(x_1, \dots, x_n) : x_i \in \mathbb{R}\}$$

is a vector space over $\mathbb{R}$, with coordinatewise addition:

$$(x_1, \dots, x_n) + (y_1, \dots, y_n) = (x_1 + y_1, \dots, x_n + y_n),$$

and coordinatewise scalar multiplication:

$$\lambda(x_1, \dots, x_n) = (\lambda x_1, \dots, \lambda x_n).$$

More examples

- $\mathbb{C}^n$ over $\mathbb{C}$.
- $\mathbb{C}$ over $\mathbb{R}$.
- $\mathbb{R}[x]$, the set of real polynomials, over $\mathbb{R}$.
- $M_2(\mathbb{Z}_p)$ over $\mathbb{Z}_p$, where p is prime.

For matrices over $\mathbb{Z}_p$,

$$\lambda \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} \lambda a & \lambda b \\ \lambda c & \lambda d \end{pmatrix},$$

with entries computed modulo p .

Vector spaces over finite fields

Because $\mathbb{Z}_p$ is a field when p is prime, we can form vector spaces over $\mathbb{Z}_p$.

For example,

$$\mathbb{Z}_2^n = \{0, 1\}^n$$

is a vector space over $\mathbb{Z}_2$.

Addition is coordinatewise modulo 2:

$$(1, 0, 1) + (1, 1, 0) = (0, 1, 1).$$

This viewpoint will be important later for linear error-correcting codes.

Subspaces

Let V be a vector space over a field F , and let W be a nonempty subset of V :

$$W \subseteq V.$$

We say that W is a **subspace** of V if W is a vector space using the same addition and scalar multiplication as V .

Equivalently, $W \neq \emptyset$ is a subspace if it satisfies the following two closure properties:

$$u, v \in W \Rightarrow u + v \in W, \quad \lambda \in F, v \in W \Rightarrow \lambda v \in W.$$

Example in $\mathbb{R}^2$:

$$W = \{(t, 2t) : t \in \mathbb{R}\}$$

is a subspace, but

$$W' = \{(t, 2t + 1) : t \in \mathbb{R}\}$$

is not, since it does not contain $(0, 0)$.

Span and generators

Given vectors $v_1, \dots, v_k \in V$, their span is

$$\text{span}(v_1, \dots, v_k) = \{\lambda_1 v_1 + \dots + \lambda_k v_k : \lambda_i \in F\}.$$

This is the smallest subspace containing $v_1, \dots, v_k$.

Analogy:

$\text{span}(v_1, \dots, v_k)$ in vector spaces

corresponds to

$\langle g_1, \dots, g_k \rangle$ in groups.

The group version is often much harder.

Quick check: vector spaces

① Is $\mathbb{Z}_6^2$ a vector space over $\mathbb{Z}_6$?

② Is $\mathbb{Z}_5^2$ a vector space over $\mathbb{Z}_5$?

③ Is

$$W = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}$$

a subspace of $\mathbb{R}^3$?

④ Is

$$U = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 1\}$$

a subspace of $\mathbb{R}^3$?

Group preliminaries

Why work abstractly?

We have many groups:

$$\mathbb{Z}, \quad \mathbb{Z}_n, \quad G_n, \quad S_n, \quad A_n, \quad D_n, \quad \mathrm{GL}(n, \mathbb{R}).$$

Instead of proving the same fact separately in each example, we prove it once from the group axioms.

Then the result applies to every group.

This is one of the main advantages of abstraction.

Solving equations and cancellation

Let G be a group and let $a, b \in G$.

The equation $a = bx$ has the unique solution

$$x = b^{-1}a.$$

The equation $a = yb$ has the unique solution

$$y = ab^{-1}.$$

Hence we may cancel on the same side:

$$ba = bc \Rightarrow a = c, \quad ab = cb \Rightarrow a = c.$$

But the two equations $a = bx$ and $a = yb$ generally have different solutions in a non-Abelian group.

Warning: order matters

In a non-Abelian group, multiplication order matters.

From

$$ab = ac$$

we may cancel a on the left and conclude $b = c$.

From

$$ba = ca$$

we may cancel a on the right and conclude $b = c$.

But from

$$ab = ca$$

we cannot simply cancel a .

Example equation:

$$xaba^{-1} = c \implies x = cab^{-1}a^{-1}.$$

Powers in a group

Let $g \in G$.

Define

$$g^1 = g, \quad g^{k+1} = gg^k, \quad g^0 = e, \quad g^{-k} = (g^{-1})^k.$$

If the group operation is written additively, then powers become multiples:

$$g^n \leftrightarrow ng.$$

Example in $(\mathbb{Z}, +)$:

$$\langle 3 \rangle = \{3n : n \in \mathbb{Z}\}.$$

Index laws

For all integers r, s :

$$g^r g^s = g^{r+s},$$

$$(g^r)^s = g^{rs},$$

$$g^{-r} = (g^r)^{-1} = (g^{-1})^r.$$

If g and h commute, then

$$(gh)^r = g^r h^r.$$

Without commutativity, this last formula may fail:

$$(gh)^2 = ghgh \neq g^2 h^2$$

in general.

Order of an element

Let $g \in G$.

If there is no positive integer n such that

$$g^n = e,$$

then g has **infinite order**.

Otherwise, the **order** of g is the smallest positive integer n such that

$$g^n = e.$$

Notation:

$$\text{ord}(g) = n.$$

Examples of element orders

In S_3 :

$$\text{ord}((1\ 2)) = 2, \quad \text{ord}((1\ 2\ 3)) = 3.$$

In $(\mathbb{Z}_{12}, +)$:

$$\text{ord}([4]_{12}) = 3, \quad 3[4]_{12} = [0]_{12}.$$

In $\text{GL}(2, \mathbb{R})$,

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \Rightarrow A^n = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}.$$

So A has infinite order over $\mathbb{R}$, but order p over $\mathbb{Z}_p$.

Periodicity of powers

Theorem. Suppose $g \in G$ has finite order n . Then

$$g^r = g^s$$

if and only if

$$r \equiv s \pmod{n}.$$

Consequences:

- the distinct powers are

$$e, g, g^2, \dots, g^{n-1};$$

- after that, the pattern repeats;
- the order of g is the number of distinct powers of g .

Subgroups

Definition. Let G be a group.

A non-empty subset $H \subseteq G$ is a **subgroup** of G if H is itself a group under the same operation as G .

Notation:

$$H \leq G.$$

Every group has two obvious subgroups:

$$\{e\} \leq G, \quad G \leq G.$$

These are the **trivial subgroup** and the **whole group**.

What a subgroup must contain

If $H \leq G$, then:

- H contains the identity element e of G ;
- if $h \in H$, then $h^{-1} \in H$;
- if $h, k \in H$, then $hk \in H$.

Important point:

The identity and inverses in H are the same as the identity and inverses in G .

The subgroup test

Let H be a non-empty subset of a group G .

Then the following are equivalent:

- ① H is a subgroup of G .
- ② H is closed under products and inverses:

$$h, k \in H \Rightarrow hk \in H, \quad h \in H \Rightarrow h^{-1} \in H.$$

- ③ For all $h, k \in H$,

$$hk^{-1} \in H.$$

The third version is often the fastest.

Proof idea for the subgroup test

Assume H is non-empty and

$$h, k \in H \Rightarrow hk^{-1} \in H.$$

Identity: take $h = k$:

$$hh^{-1} = e \in H.$$

Inverses: since $e \in H$ and $g \in H$,

$$eg^{-1} = g^{-1} \in H.$$

Products: if $x, y \in H$, then $y^{-1} \in H$, so

$$x(y^{-1})^{-1} = xy \in H.$$

Associativity is inherited from G .

Subgroup test in additive notation

For an additive group, inverses are negatives.

The subgroup test becomes:

A non-empty subset H of an additive group G is a subgroup if

$$h, k \in H \Rightarrow h - k \in H.$$

Example:

$$2\mathbb{Z} = \{2m : m \in \mathbb{Z}\} \leq (\mathbb{Z}, +),$$

because

$$2a - 2b = 2(a - b) \in 2\mathbb{Z}.$$

Examples of subgroups

- $2\mathbb{Z} \leq (\mathbb{Z}, +)$.
- $(\mathbb{Z}, +) \leq (\mathbb{R}, +) \leq (\mathbb{C}, +)$.
- $A_n \leq S_n$.
- The invertible diagonal matrices form a subgroup of $GL(n, \mathbb{R})$.
- The determinant-one matrices form a subgroup:

$$SL(n, \mathbb{R}) = \{A \in GL(n, \mathbb{R}) : \det A = 1\}.$$

For $SL(n, \mathbb{R})$:

$$\det(AB) = \det(A) \det(B), \quad \det(A^{-1}) = \frac{1}{\det(A)}.$$

Non-examples of subgroups

- 1 $\mathbb{N}$ is not a subgroup of $(\mathbb{Z}, +)$.

It is not closed under additive inverses.

- 2 $\mathbb{R} \setminus \{0\}$ is not a subgroup of $(\mathbb{R}, +)$.

It does not contain the identity 0, and it is not closed under addition:

$$1 + (-1) = 0.$$

- 3 The odd permutations are not a subgroup of S_n .

The product of two odd permutations is even.

Intersections of subgroups

Theorem. If $H \leq G$ and $K \leq G$, then

$$H \cap K \leq G.$$

Proof using the subgroup test. Since both H and K contain e , the intersection is non-empty.

Take $x, y \in H \cap K$. Then

$$xy^{-1} \in H \quad \text{and} \quad xy^{-1} \in K.$$

Hence

$$xy^{-1} \in H \cap K.$$

Example in $(\mathbb{Z}, +)$:

$$\langle 2 \rangle \cap \langle 3 \rangle = 2\mathbb{Z} \cap 3\mathbb{Z} = 6\mathbb{Z} = \langle 6 \rangle.$$

The subgroup generated by one element

Let G be a group and let $g \in G$.

Define

$$\langle g \rangle = \{g^n : n \in \mathbb{Z}\}.$$

This is the **subgroup generated by g** .

Why is it a subgroup? If

$$h = g^i, \quad k = g^j,$$

then

$$hk^{-1} = g^i(g^j)^{-1} = g^{i-j} \in \langle g \rangle.$$

If $\text{ord}(g) = n$, then

$$\langle g \rangle = \{e, g, g^2, \dots, g^{n-1}\}.$$

Cyclic groups and S_3

A group G is **cyclic** if

$$G = \langle g \rangle$$

for some $g \in G$.

Every cyclic group is Abelian.

Cyclic subgroups of S_3 :

$$\langle \text{id} \rangle = \{\text{id}\},$$

$$\langle (1\ 2) \rangle = \{\text{id}, (1\ 2)\}, \quad \langle (1\ 3) \rangle = \{\text{id}, (1\ 3)\}, \quad \langle (2\ 3) \rangle = \{\text{id}, (2\ 3)\},$$

$$\langle (1\ 2\ 3) \rangle = \{\text{id}, (1\ 2\ 3), (1\ 3\ 2)\}.$$

So S_3 is not cyclic.

Cyclic subgroups of $\mathbb{Z}_{12}$ and more generators

In additive notation,

$$\langle [a]_{12} \rangle = \{n[a]_{12} : n \in \mathbb{Z}\}.$$

Examples:

$$\langle [1] \rangle = \mathbb{Z}_{12}, \quad \langle [2] \rangle = \{[0], [2], [4], [6], [8], [10]\},$$

$$\langle [3] \rangle = \{[0], [3], [6], [9]\}, \quad \langle [4] \rangle = \{[0], [4], [8]\}.$$

Groups may also be generated by several elements:

$$D_n = \langle \rho, R \rangle, \quad S_n = \langle (1\ 2), (2\ 3), \dots, (n-1\ n) \rangle.$$

Final quick checks and summary

Quick checks.

- 1 Is $3\mathbb{Z}$ a subgroup of $(\mathbb{Z}, +)$?
- 2 Is $\{\text{id}, (1\ 2), (1\ 3), (2\ 3)\}$ a subgroup of S_3 ?
- 3 Find $\langle [4]_{10} \rangle$ in $(\mathbb{Z}_{10}, +)$. What is the order of $[4]_{10}$?
- 4 Is A_3 cyclic?

Key takeaways.

- Fields are rings where division by non-zero elements is possible.
- Vector spaces are Abelian groups with scalar multiplication by a field.
- The subgroup test is the central tool of Section 5.1.
- Cyclic subgroups are generated by powers of one element.

Next step: cosets and Lagrange's Theorem.